

AS3935 LIGHTNING DETECTOR

Antenna Calibration Report

Station ID	GLENCORE WONDERKOP
Sensor	AS3935
Interface	SPI bus 0, device 0, mode 1, 1 MHz; IRQ/LCO on GPIO17
Calibration date	5/28/2026

1. Reason for Calibration

The AS3935 uses a resonant LC antenna that must be tuned to 500 kHz +/- 3.5% for correct lightning energy and distance estimation. The sensor provides 16 internal tuning-capacitor steps (0 to 120 pF, in 8 pF increments).

2. Method

The antenna LC oscillator (LCO) is routed to the IRQ pin in DISP_LCO mode. To allow accurate software edge-counting on the Pi, the AS3935 internal /128 frequency divider is enabled, bringing the ~500 kHz signal down to ~3.9 kHz at the pin. Each of the 16 tuning-capacitor settings is applied in turn; rising edges are counted over a 1.000 s window and multiplied back by 128 to recover the true antenna frequency. Error is the absolute difference from 500,000 Hz, expressed as a percentage. Acceptable antenna band: 482,500 Hz to 517,500 Hz.

3. Measured Results

Cap step	Cap (pF)	Measured freq (Hz)	Error (%)
0	0	519,040	3.81
1	8	516,608	3.32
2	16	515,584	3.12
3	24	513,536	2.71
4	32	511,872	2.37
5	40	510,080	2.02
6	48	508,288	1.66
7	56	507,008	1.40
8	64	505,216	1.04
9	72	503,808	0.76
10	80	501,632	0.33
11	88	500,608	0.12
12	96	498,816	0.24
13	104	497,152	0.57
14	112	496,000	0.80
15	120	494,464	1.11

4. Selected Setting

Tune Cap	11
Tuning Capacitance	88 pF
Measured Frequency	500,608 Hz
Frequency Error	0.12 % (target +/- 3.5 %)
Result	PASS

The selected setting is the lowest-error step and sits essentially on the 500 kHz target, with roughly 29x margin against the +/- 3.5% tolerance. 15 of the 16 steps fall inside tolerance, confirming there is no hardware fault. This value is stored in the unit configuration as tune_cap = 11.

5. Configured Detection Parameters

Setting	Value	Notes
AFE / gain mode	OUTDOOR	
Noise floor	5	Raise if site EMI causes false noise events
Watchdog threshold	5	
Spike rejection	5	
Minimum strikes	5	Strikes before a validated event
Mask disturbers	Enabled	Rejects man-made interference

6. Calibration Notes

This calibration reflects the unit's current (pre-installation) position. The antenna's resonant frequency is sensitive to nearby grounded metalwork and to ambient temperature. Calibration should be performed after installation as well.